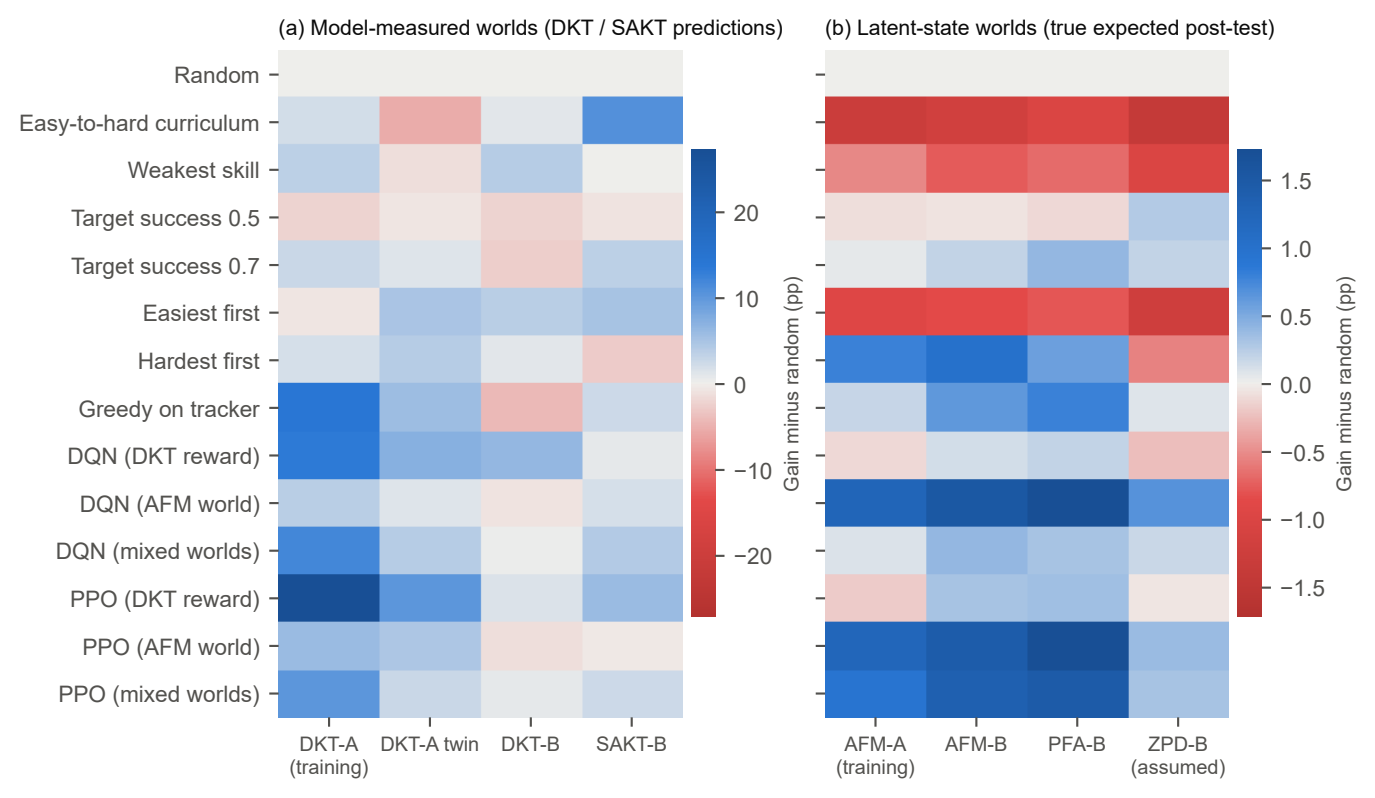
(a) Model-measured worlds (DKT / SAKT predictions)
(b) Latent-state worlds (true expected post-test)
Random
Easy-to-hard curriculum
Weakest skill
Target success 0.5
Target success 0.7
Easiest first
Hardest first
Greedy on tracker
DQN (DKT reward)
DQN (AFM world)
DQN (mixed worlds)
PPO (DKT reward)
PPO (AFM world)
PPO (mixed worlds)
DKT-A (training)
DKT-A twin
DKT-B
SAKT-B
AFM-A (training)
AFM-B
PFA-B
ZPD-B (assumed)
Gain minus random (pp)
20
10
0
−10
−20
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5